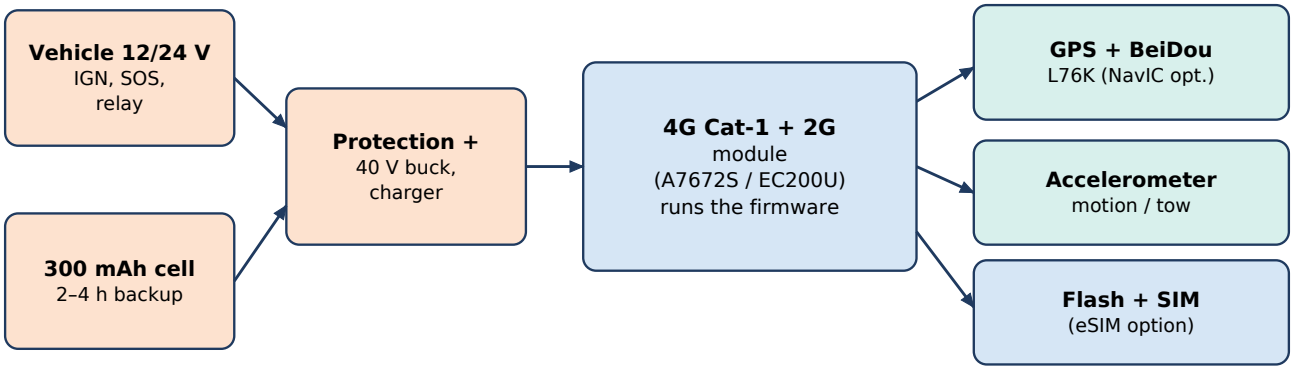
Samay Track Lite: product and cost

A low-cost 4G GPS tracker for bikes and scooters, made to sell on Amazon, through dealers and to small fleets. It is a **new design**: a 2-layer board where one 4G module also runs the firmware. All figures are estimates (Oct 2026, ex-GST).

Spec	Lite
Network	4G LTE Cat-1 **+ 2G fallback** (SIMCom A7672S / Quectel EC200U)
GPS	GPS + BeiDou (L76K class); NavIC chip optional (+₹300–400)
Features	live tracking · theft / tow alert · engine lock · geofence · ignition · SOS input · power-cut alert
Battery	300 mAh internal, ≈ 2–4 h after a power cut
Supply	9–32 V (12 / 24 V vehicles), reverse protected
Size / box	≈ 50 × 35 mm board, IP65 plastic box, 4-wire harness
Not included	NavIC dual-band, 40 h battery, crash detection, AIS-140

Design



Block diagram: the 4G module is also the processor, so no ESP32 is needed.

Cost per unit (1,000 pcs)

Part	₹
4G Cat-1 + 2G module	500–650
GPS module	200–250
Accelerometer, power, charger, 300 mAh cell	140–190
Antennas (GPS patch + LTE FPC)	60–80
Flash, SIM holder, ESD, passives	80–100
2-layer PCB + assembly	80–100
Box, harness, packaging	130–170
Total	**≈ 1,200–1,500**

Price and profit

Assumed hardware cost ₹1,350. Each unit also carries ₹400 year-1 data and cloud, ₹100 shipping and a 3 % warranty reserve. Self-install, so there is no installation cost.

Channel	Price	Fees	Profit / unit	Margin
Own website / WhatsApp	₹2,999	—	₹1,059	35 %
Own website / WhatsApp	₹3,499	—	₹1,544	44 %
Dealer (18 % cut)	₹3,499	18 %	₹914	26 %
Amazon (18 % fees + 7 % ads)	₹2,999	25 %	₹309	10 %
Amazon (18 % fees + 7 % ads)	₹3,499	25 %	₹669	19 %
Small fleet / EV maker, bulk	₹2,499	—	₹574	23 %
Renewal / year	₹999	—	₹599	60 %

Recommendation: list at **₹3,499 including the first year** (₹999/yr after). At ₹2,999 on Amazon, only about ₹300 per unit is left after fees and ads. Sell direct and to fleets for profit, and use Amazon for reach.

Competition at this price

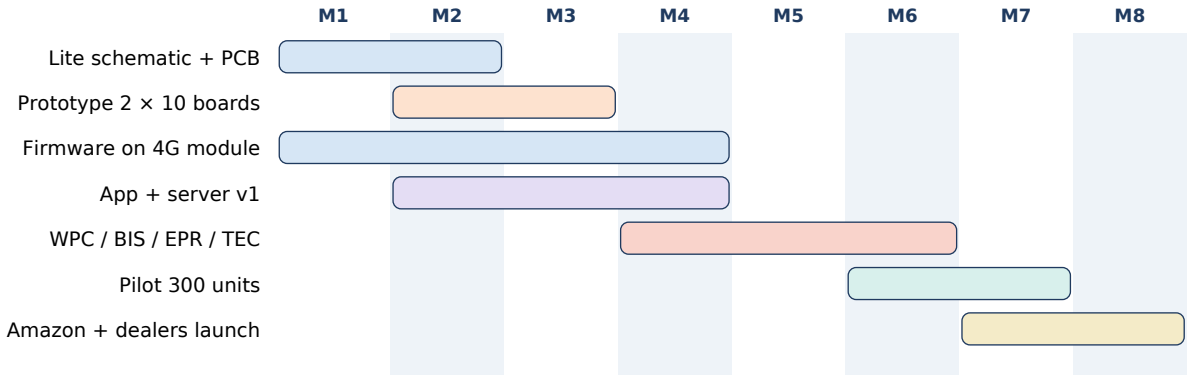
Product	Price
SeTrack, Ajjas 4G Lite, Onelap Micro / LocoShield, Fleettrack 4G	₹900–2,900 incl. year 1
Alike GPS VT05R	from ₹3,999

Our edge at ₹3,499: 2G fallback and normal 4G coverage, an engine lock with safe-stop logic, a power-cut alert with backup, a clean app, and local support. Optional NavIC is a selling point that the cheap ones lack.

Money needed and timeline

Item	Low (₹ lakh)	High (₹ lakh)
Lite board design + external review	0.5	1
Prototypes (2 rounds × 10)	0.5	1
Firmware + app + server	2	5
Enclosure: stock IP65 box → custom mould later	0.3	4
Test fixture	0.3	0.8
Certificates: WPC, BIS (certified cell), EPR, TEC, legal	3	8
Pilot batch 300 units × ≈ ₹1,800	5.4	5.4
Launch: Amazon listing, video, ads	1	3
6 months running costs (cloud, SIM, support)	3	6
Total to launch	**16**	**34.2**

Then about **₹12–15 lakh per 1,000 units** of stock. **Break-even:** with 30 % direct, 40 % Amazon and 30 % dealer sales at ₹3,499, the average profit is ≈ ₹1,005 per unit, so ≈ **2,500 units** recover ₹25 lakh. Renewals at ≈ ₹600 per customer per year shorten that.



About 8 months from start to launch.

Still missing: decide or do these

#	Item
1	Choose the module: A7672S or EC200U (price, SDK, supplier in India)
2	Lite schematic + PCB (Claude can design it, same flow as the full board)
3	Firmware on the 4G module + the app/server (Claude can write them; you create the accounts)
4	IoT SIM deal (Jio / Airtel) and DoT M2M registration if you bundle the data
5	Trademark check for "Samay", GST, Amazon seller account
6	Warranty, returns, and a self-install video